

Data →

↓
characterize
(measure scales)

Statistical tools

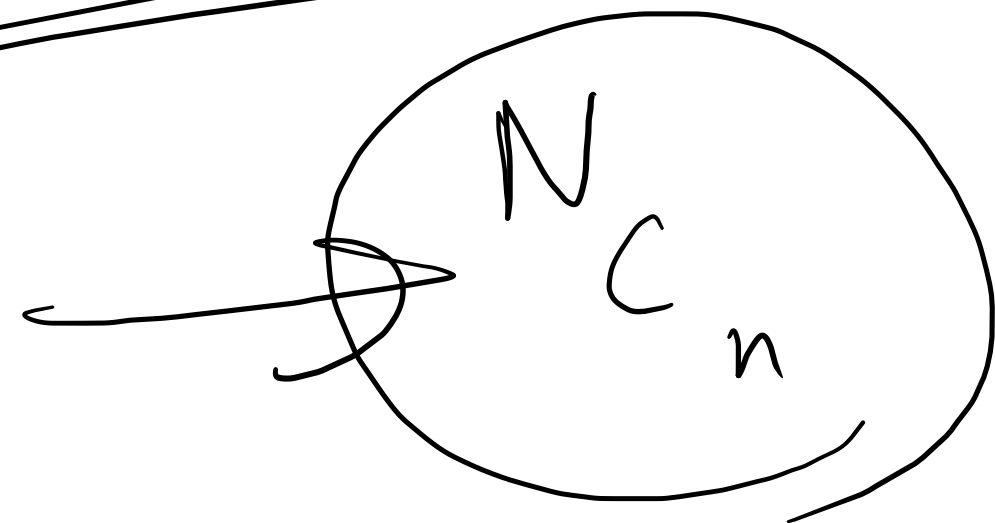
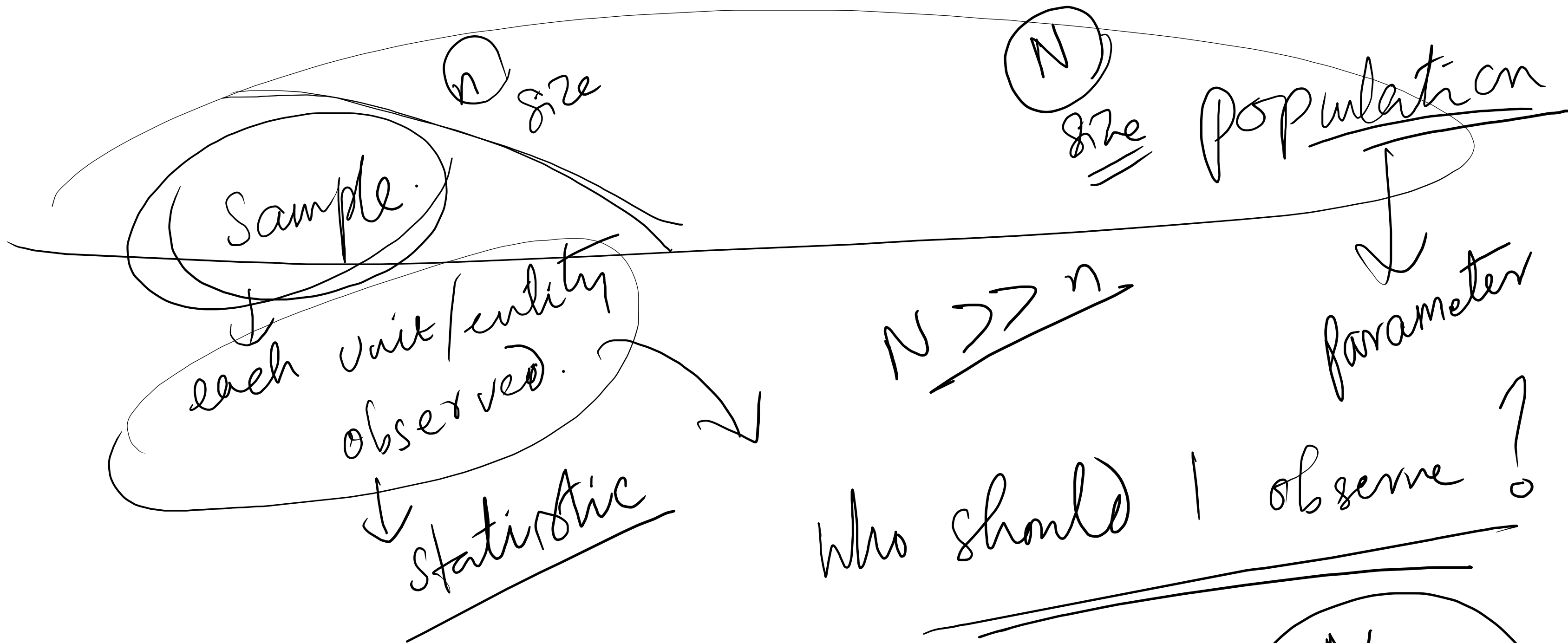
Expt

eg Effect of Vaccination

↓
Sampling

Indian population

target



methods of sampling

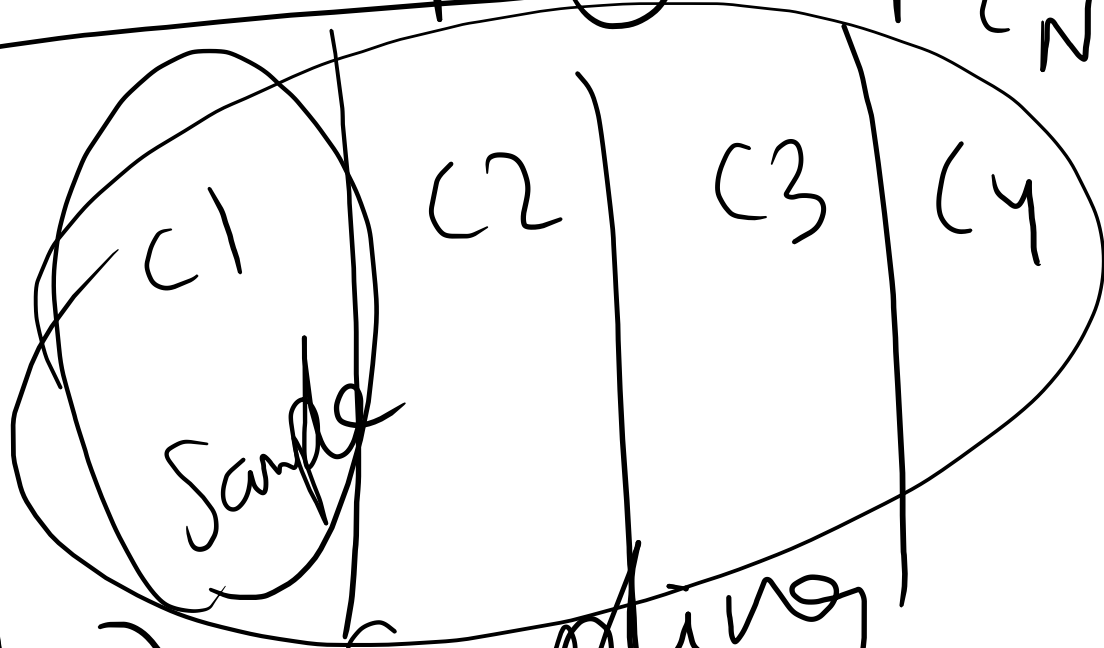
- Simple Random Sampling

N
 C_N

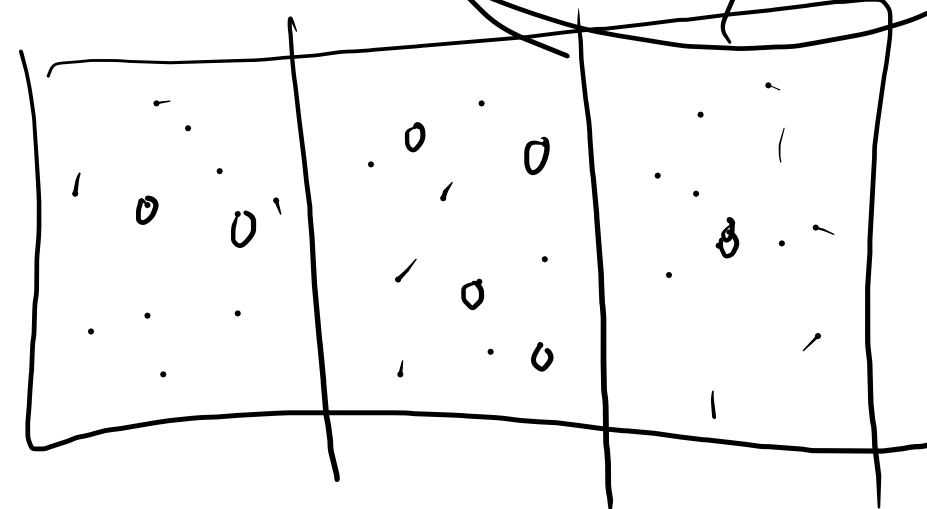
1, 2, 3, 4

Cluster Sampling

N
 C_N



- Stratified Sampling



2 points

1, 2
1, 3
1, 4

Expt Design

Sampling Bias



Problem Defⁿ

Design of Expts

Research Question

target

Response var

Factors / levels

Experimental unit

Choose Design



Randomization Replication control/Block

Collect data

↓
IDA

↓
EDA

Is the Data reliable?

What patterns are there?